SIMATS ENGINE ERING
ASSESSMENT TOOL 3

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO1: *Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)*

QUESTIONS: 05 **Total Marks:100**

Annexure A

Exploratory Data Analysis - Based Practical Assessment

Code of Conduct:

I M.Nandhini(192472318) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M.Nandhini

Annexure A

Exploratory Data Analysis-Based Practical Assessment

1. Education Analytics Dataset

Question 1

A college wants to analyze student performance based on attendance, study hours, internal marks, and final result. Perform EDA to identify the factors affecting student results.

Student_ID	Attendance	Study_Hours	Internal_Marks	Result
S01	92	5	45	Pass
S02	55	2	24	Fail
S03	78	4	38	Pass
S04	60	3	28	Fail
S05	88	5	42	Pass
S06	47	1	20	Fail
S07	95	6	48	Pass
S08	72	3	35	Pass
S09	50	2	22	Fail
S10	83	4	40	Pass
S11	66	3	30	Fail
S12	90	5	44	Pass
S13	58	2	25	Fail
S14	76	4	36	Pass
S15	62	3	29	Fail
S16	85	5	41	Pass
S17	45	1	18	Fail
S18	80	4	39	Pass
S19	68	3	32	Pass
S20	53	2	23	Fail

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
data = {
    'Attendance': [92,55,78,60,88,47,95,72,50,83,66,90,58,76,62,85,45,80,68,53],
    'Study_Hours': [5,2,4,3,5,1,6,3,2,4,3,5,2,4,3,5,1,4,3,2],
    'Internal_Marks': [45,24,38,28,42,20,48,35,22,40,30,44,25,36,29,41,18,39,32,23],
    'Result': ['Pass','Fail','Pass','Fail','Pass','Fail','Pass','Pass','Fail','Pass',
               'Fail','Pass','Fail','Pass','Fail','Pass','Fail','Pass','Pass','Fail']
}
df = pd.DataFrame(data)
print("Dataset Shape:", df.shape)
print("\nData Types:\n", df.dtypes)
print("\nMissing Values:\n", df.isnull().sum())
print("\nDuplicate Rows:", df.duplicated().sum())
print("\nStatistical Summary:\n", df.describe())
print("\nResult Count:\n", df['Result'].value_counts())
sns.countplot(data=df, x='Result')
plt.title("Student Results")
plt.show()
df['Attendance'].hist()
plt.title("Attendance Distribution")
plt.xlabel("Attendance")
plt.ylabel("Number of Students")
plt.show()
sns.scatterplot(data=df, x='Study_Hours',
                y='Internal_Marks', hue='Result')
plt.title("Study Hours vs Internal Marks")
plt.show()
sns.boxplot(data=df, x='Result', y='Attendance')
plt.title("Attendance by Result")
plt.show()
```

```
Dataset Shape: (20, 4)

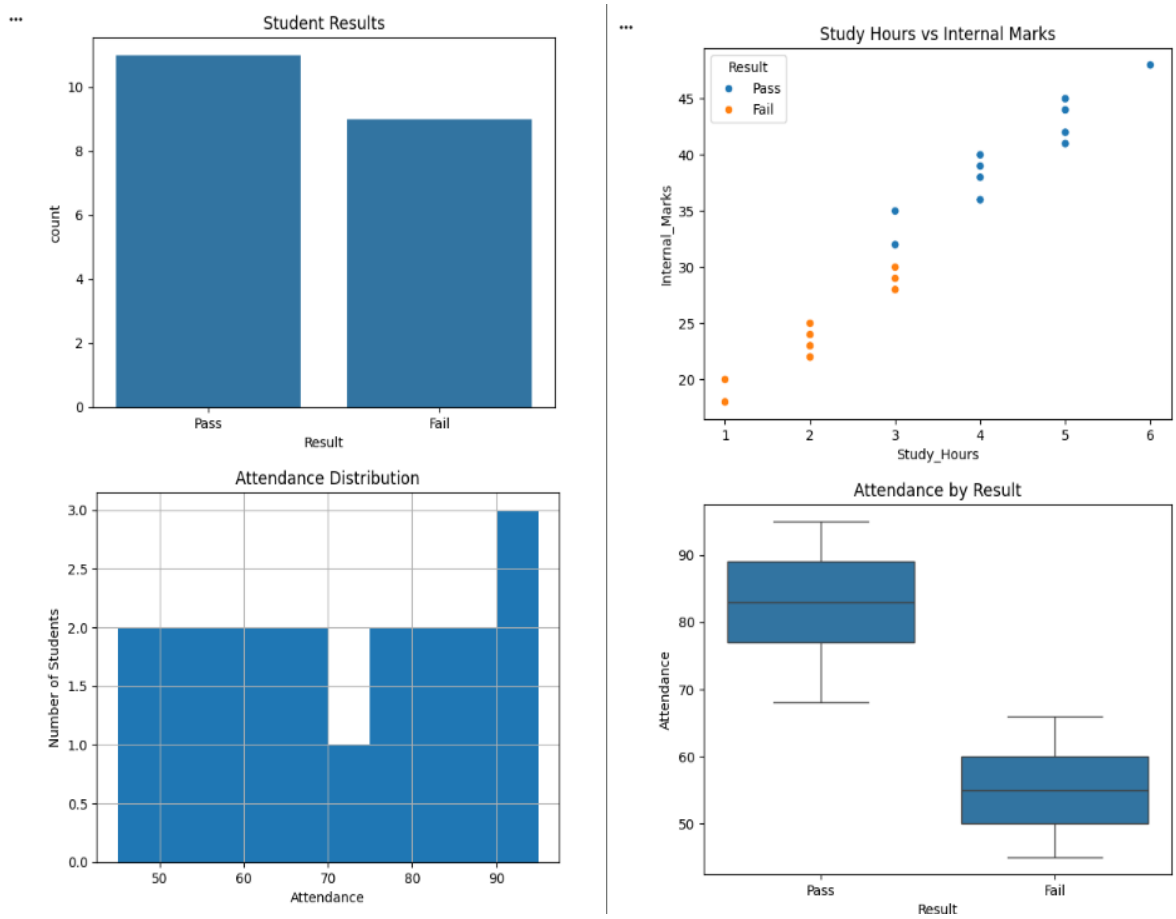
Data Types:
 Attendance         int64
Study_Hours         int64
Internal_Marks      int64
Result             object
dtype: object

Missing Values:
 Attendance        0
Study_Hours        0
Internal_Marks     0
Result             0
dtype: int64

Duplicate Rows: 0

Statistical Summary:
       Attendance  Study_Hours  Internal_Marks
count   20.000000    20.000000       20.000000
mean    70.150000     3.350000       32.950000
std     15.938286     1.424411        9.087151
min     45.000000     1.000000       18.000000
25%     57.250000     2.000000       24.750000
50%     70.000000     3.000000       33.500000
75%     83.500000     4.250000       40.250000
max     95.000000     6.000000       48.000000

Result Count:
 Result
Pass    11
Fail     9
Name: count, dtype: int64
```



2. Healthcare Analytics Dataset

Question 2

A hospital wants to analyze patient health risk based on age, blood sugar, blood pressure, BMI, and disease status. Perform EDA to find important health patterns.

Patient_ID	Age	Sugar_Level	BP	BMI	Disease_Status
P01	45	145	130	28	Yes
P02	32	98	118	23	No
P03	55	180	145	31	Yes
P04	28	90	110	22	No
P05	60	200	150	34	Yes
P06	40	120	125	26	No
P07	52	165	140	30	Yes
P08	35	105	120	24	No
P09	48	155	135	29	Yes
P10	30	95	115	23	No
P11	62	210	155	35	Yes
P12	38	115	122	25	No
P13	50	170	142	30	Yes
P14	27	88	108	21	No
P15	58	190	148	33	Yes
P16	42	130	128	27	No
P17	46	150	132	28	Yes
P18	34	100	116	24	No
P19	53	175	144	31	Yes

P20	29	92	112	22	No

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
data = {
    'Age': [45,32,55,28,60,40,52,35,48,30,62,38,50,27,58,42,46,34,53,29],
    'Sugar_Level': [145,98,180,90,200,120,165,105,155,95,210,115,170,88,190,130,150,100,175,92],
    'BP': [130,118,145,110,150,125,140,120,135,115,155,122,142,108,148,128,132,116,144,112],
    'BMI': [28,23,31,22,34,26,30,24,29,23,35,25,30,21,33,27,28,24,31,22],
    'Disease_Status': ['Yes','No','Yes','No','Yes','No','Yes','No','Yes','No',
                       'Yes','No','Yes','No','Yes','No','Yes','No','Yes','No']
}

df = pd.DataFrame(data)
print("Dataset Shape:", df.shape)
print("\nData Types:\n", df.dtypes)
print("\nMissing Values:\n", df.isnull().sum())
print("\nDuplicate Rows:", df.duplicated().sum())
print("\nStatistical Summary:\n", df.describe())
print("\nDisease Status Count:\n", df['Disease_Status'].value_counts())
sns.countplot(data=df, x='Disease_Status')
plt.title("Disease Status Distribution")
plt.show()
df['Sugar_Level'].hist()
plt.title("Blood Sugar Level Distribution")
plt.xlabel("Sugar Level")
plt.ylabel("Number of Patients")
plt.show()
sns.scatterplot(data=df, x='Age', y='Sugar_Level',
                hue='Disease_Status')
plt.title("Age vs Sugar Level")
plt.show()
sns.boxplot(data=df, x='Disease_Status', y='BMI')
plt.title("BMI by Disease Status")
plt.show()
```

```
Dataset Shape: (20, 5)

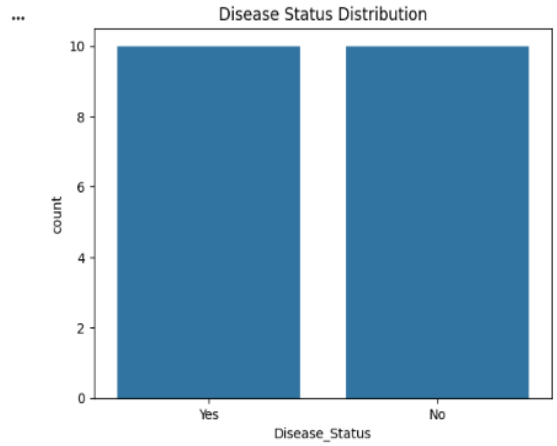
Data Types:
 Age              int64
Sugar_Level       int64
BP                int64
BMI               int64
Disease_Status    object
dtype: object

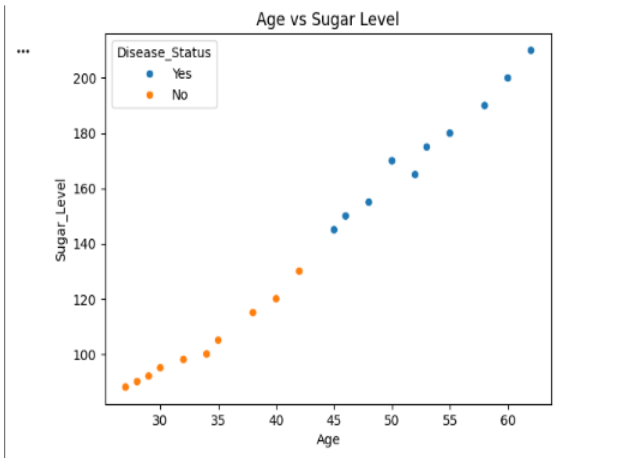
Missing Values:
 Age              0
Sugar_Level       0
BP                0
BMI               0
Disease_Status    0
dtype: int64

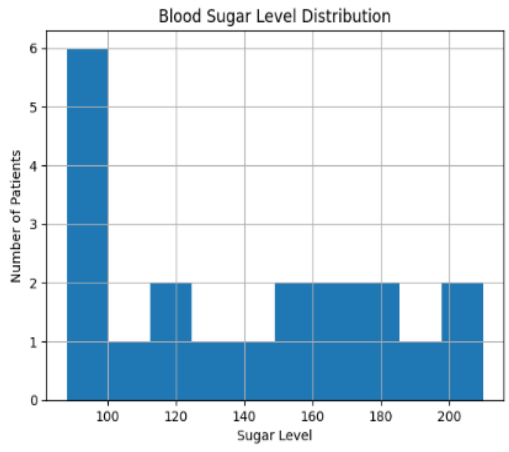
Duplicate Rows: 0

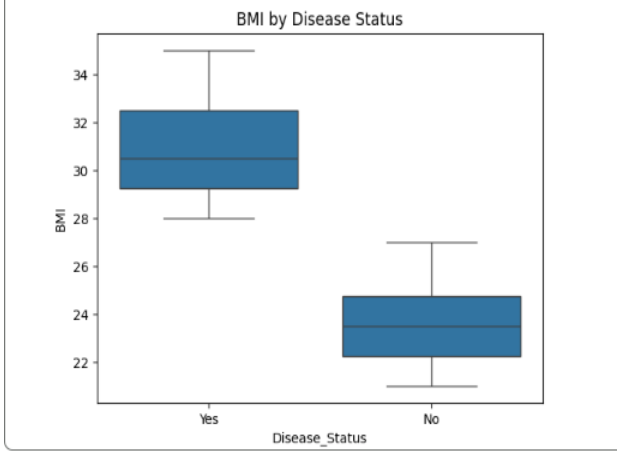
Statistical Summary:
              Age  Sugar_Level          BP        BMI
count  20.000000    20.000000   20.000000  20.000000
mean   43.200000   138.650000  129.750000  27.300000
std    11.316499    40.334491   14.523574   4.256265
min    27.000000    88.000000  108.000000  21.000000
25%    33.500000    99.500000  117.500000  23.750000
50%    43.500000   137.500000  129.000000  27.500000
75%    52.250000   171.250000  142.500000  30.250000
max    62.000000   210.000000  155.000000  35.000000

Disease Status Count:
 Disease_Status
Yes    10
No     10
Name: count, dtype: int64
```









3. Retail Sales Dataset

Question 3

A supermarket wants to analyze product sales based on category, price, quantity sold, discount, and revenue. Perform EDA to identify high-selling products and revenue patterns.

Product_ID	Category	Price	Quantity_Sold	Discount	Revenue
PR01	Grocery	50	20	5	1000
PR02	Snacks	30	35	3	1050
PR03	Dairy	45	18	2	810
PR04	Fruits	60	25	5	1500
PR05	Vegetables	40	30	4	1200
PR06	Grocery	80	15	6	1200
PR07	Snacks	25	40	3	1000
PR08	Dairy	55	16	2	880
PR09	Fruits	70	22	5	1540
PR10	Vegetables	35	28	4	980
PR11	Grocery	90	12	6	1080
PR12	Snacks	20	45	2	900
PR13	Dairy	65	14	3	910
PR14	Fruits	75	20	5	1500
PR15	Vegetables	45	26	4	1170
PR16	Grocery	100	10	7	1000
PR17	Snacks	35	32	3	1120
PR18	Dairy	50	19	2	950
PR19	Fruits	80	18	6	1440
PR20	Vegetables	30	35	3	1050

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
data = {
    'Category': ['Grocery','Snacks','Dairy','Fruits','Vegetables',
                 'Grocery','Snacks','Dairy','Fruits','Vegetables',
                 'Grocery','Snacks','Dairy','Fruits','Vegetables',
                 'Grocery','Snacks','Dairy','Fruits','Vegetables'],
    'Price': [50,30,45,60,40,80,25,55,70,35,90,20,65,75,45,100,35,50,80,30],
    'Quantity_Sold': [20,35,18,25,30,15,40,16,22,28,12,45,14,20,26,10,32,19,18,35],
    'Discount': [5,3,2,5,4,6,3,2,5,4,6,2,3,5,4,7,3,2,6,3],
    'Revenue': [1000,1050,810,1500,1200,1200,1000,880,1540,980,
                1080,900,910,1500,1170,1000,1120,950,1440,1050]
}
df = pd.DataFrame(data)
print("Dataset Shape:", df.shape)
print("\nData Types:\n", df.dtypes)
print("\nMissing Values:\n", df.isnull().sum())
print("\nDuplicate Rows:", df.duplicated().sum())
print("\nStatistical Summary:\n", df.describe())
print("\nCategory Count:\n", df['Category'].value_counts())
sns.barplot(data=df, x='Category', y='Revenue', estimator='sum')
plt.title("Total Revenue by Category")
plt.xticks(rotation=20)
plt.show()
df['Quantity_Sold'].hist()
plt.title("Quantity Sold Distribution")
plt.xlabel("Quantity Sold")
plt.ylabel("Number of Products")
plt.show()
sns.scatterplot(data=df, x='Quantity_Sold', y='Revenue', hue='Category')
plt.title("Quantity Sold vs Revenue")
plt.show()
sns.boxplot(data=df, x='Category', y='Revenue')
plt.title("Revenue Distribution by Category")
plt.xticks(rotation=20)
plt.show()
```

```
Dataset Shape: (20, 5)

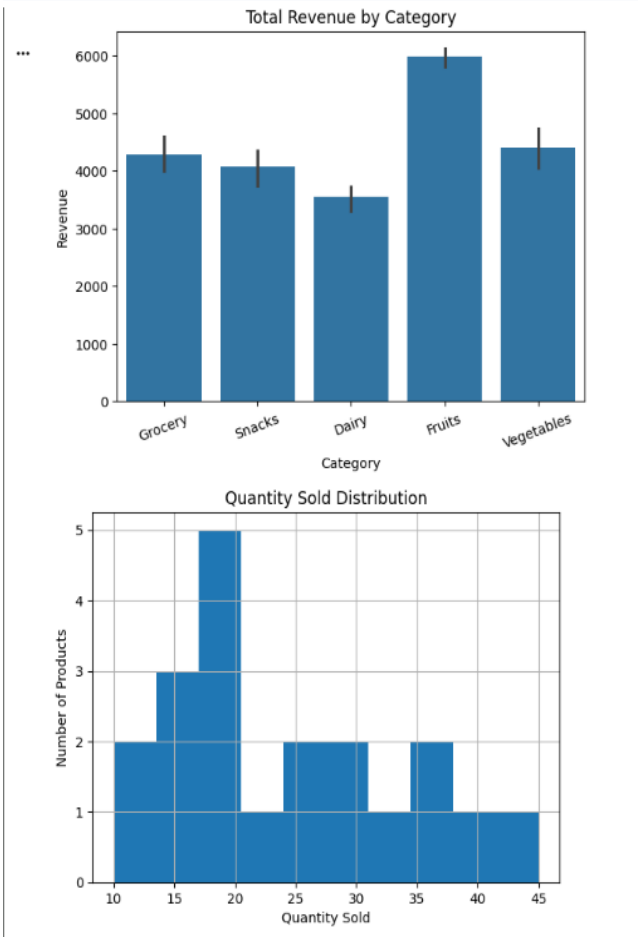
Data Types:
 Category         object
Price              int64
Quantity_Sold      int64
Discount           int64
Revenue            int64
dtype: object

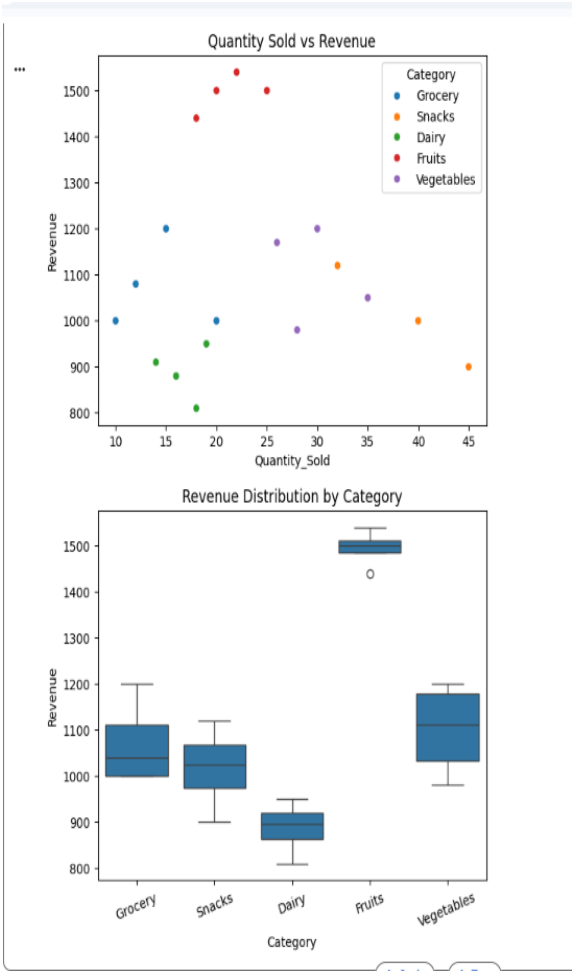
Missing Values:
 Category         0
Price             0
Quantity_Sold     0
Discount          0
Revenue           0
dtype: int64

Duplicate Rows: 0

Statistical Summary:
              Price  Quantity_Sold   Discount      Revenue
count   20.000000      20.000000  20.000000    20.000000
mean    54.000000      24.000000   4.000000  1114.000000
std     22.803509       9.673621   1.555973   221.416493
min     20.000000      10.000000   2.000000   810.000000
25%     35.000000      17.500000   3.000000   972.500000
50%     50.000000      21.000000   4.000000  1050.000000
75%     71.250000      30.500000   5.000000  1200.000000
max    100.000000      45.000000   7.000000  1540.000000

Category Count:
 Category
Grocery       4
Snacks        4
Dairy         4
Fruits        4
Vegetables    4
Name: count, dtype: int64
```





4. Banking Loan Approval Dataset

Question 4

A bank wants to analyze loan approval decisions based on income, credit score, loan amount, employment type, and approval status. Perform EDA to identify patterns related to loan approval.

Customer_ID	Income	Credit_Score	Loan_Amount	Employment_Type	Loan_Status
C01	45000	720	200000	Salaried	Approved
C02	25000	580	150000	Self-employed	Rejected
C03	60000	750	300000	Salaried	Approved
C04	22000	560	120000	Unemployed	Rejected
C05	50000	710	250000	Salaried	Approved
C06	30000	600	180000	Self-employed	Rejected
C07	70000	780	350000	Salaried	Approved
C08	28000	590	160000	Self-employed	Rejected
C09	55000	730	270000	Salaried	Approved
C10	24000	570	130000	Unemployed	Rejected
C11	65000	760	320000	Salaried	Approved
C12	35000	620	200000	Self-employed	Rejected
C13	48000	700	240000	Salaried	Approved
C14	26000	585	140000	Unemployed	Rejected
C15	75000	790	400000	Salaried	Approved
C16	32000	610	190000	Self-employed	Rejected
C17	52000	725	260000	Salaried	Approved
C18	27000	575	150000	Self-employed	Rejected

C19	68000	770	330000	Salaried	Approved
C20	23000	555	125000	Unemployed	Rejected

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
data = {
    'Income': [45000,25000,60000,22000,50000,30000,70000,28000,55000,24000,
               65000,35000,48000,26000,75000,32000,52000,27000,68000,23000],
    'Credit_Score': [720,580,750,560,710,600,780,590,730,570,
                     760,620,700,585,790,610,725,575,770,555],
    'Loan_Amount': [200000,150000,300000,120000,250000,180000,350000,160000,
                    270000,130000,320000,200000,240000,140000,400000,190000,
                    260000,150000,330000,125000],
    'Employment_Type': ['Salaried','Self-employed','Salaried','Unemployed',
                        'Salaried','Self-employed','Salaried','Self-employed',
                        'Salaried','Unemployed','Salaried','Self-employed',
                        'Salaried','Unemployed','Salaried','Self-employed',
                        'Salaried','Self-employed','Salaried','Unemployed'],
    'Loan_Status': ['Approved','Rejected','Approved','Rejected','Approved',
                    'Rejected','Approved','Rejected','Approved','Rejected',
                    'Approved','Rejected','Approved','Rejected','Approved',
                    'Rejected','Approved','Rejected','Approved','Rejected']
}
df = pd.DataFrame(data)
print("Dataset Shape:", df.shape)
print("\nData Types:\n", df.dtypes)
print("\nMissing Values:\n", df.isnull().sum())
print("\nDuplicate Rows:", df.duplicated().sum())
print("\nStatistical Summary:\n", df.describe())
print("\nLoan Status Count:\n", df['Loan_Status'].value_counts())
sns.countplot(data=df, x='Loan_Status')
plt.title("Loan Approval Status")
plt.show()
df['Credit_Score'].hist()
plt.title("Credit Score Distribution")
plt.xlabel("Credit Score")
plt.ylabel("Number of Customers")
plt.show()
sns.scatterplot(data=df, x='Income', y='Credit_Score',
                hue='Loan_Status')
plt.title("Income vs Credit Score")
plt.show()
sns.boxplot(data=df, x='Loan_Status', y='Loan_Amount')
plt.title("Loan Amount by Loan Status")
plt.show()
```

```
Dataset Shape: (20, 5)

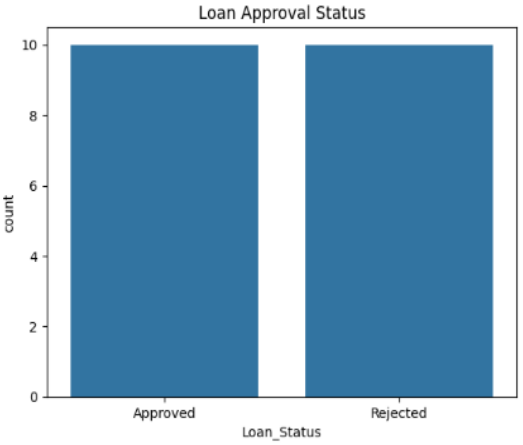
Data Types:
 Income              int64
Credit_Score         int64
Loan_Amount          int64
Employment_Type     object
Loan_Status         object
dtype: object

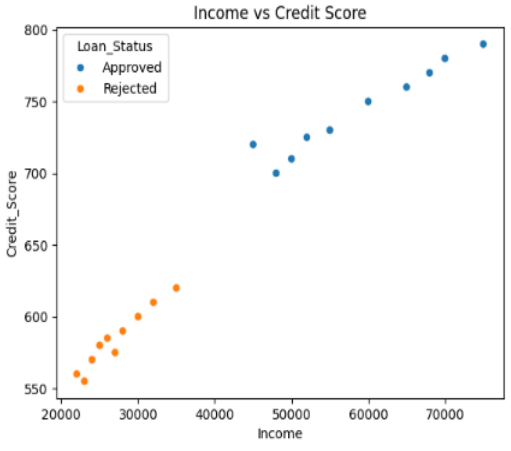
Missing Values:
 Income             0
Credit_Score        0
Loan_Amount         0
Employment_Type     0
Loan_Status         0
dtype: int64

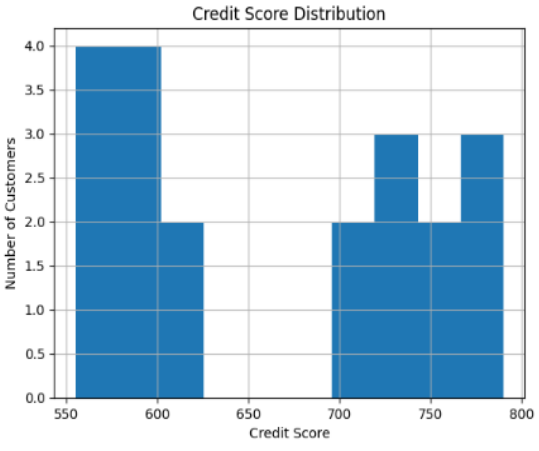
Duplicate Rows: 0

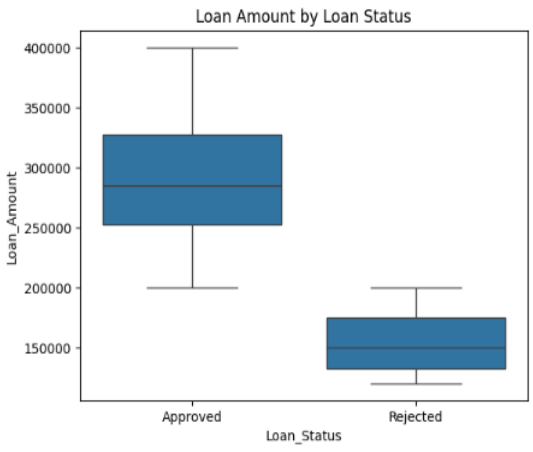
Statistical Summary:
              Income  Credit_Score    Loan_Amount
count      20.000000     20.000000      20.000000
mean    43000.000000    664.000000  223250.000000
std     17923.815383     85.526235   83733.207272
min     22000.000000    555.000000  120000.000000
25%     26750.000000    583.750000  150000.000000
50%     40000.000000    660.000000  200000.000000
75%     56250.000000    735.000000  277500.000000
max     75000.000000    790.000000  400000.000000

Loan Status Count:
 Loan_Status
Approved    10
Rejected    10
Name: count, dtype: int64
```









5. Agriculture Crop Yield Dataset

Question 5

An agricultural department wants to analyze crop yield based on rainfall, temperature, fertilizer usage, soil type, and crop yield. Perform EDA to identify the factors influencing crop production.

Farm_ID	Rainfall_mm	Temperature	Fertilizer_kg	Soil_Type	Crop_Yield_kg
F01	850	28	60	Loamy	3200
F02	600	32	45	Sandy	2100
F03	900	27	65	Loamy	3500
F04	500	34	40	Sandy	1800
F05	780	29	55	Clay	3000
F06	650	31	48	Sandy	2300
F07	920	26	70	Loamy	3700
F08	720	30	52	Clay	2800
F09	550	33	42	Sandy	1900
F10	800	28	58	Loamy	3100
F11	880	27	62	Loamy	3400
F12	620	32	46	Sandy	2200
F13	760	29	54	Clay	2950
F14	940	26	72	Loamy	3800
F15	580	33	44	Sandy	2000
F16	820	28	60	Clay	3150
F17	700	30	50	Clay	2700
F18	960	25	75	Loamy	3900
F19	540	34	41	Sandy	1850
F20	790	29	56	Clay	3050

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
data = {
    'Rainfall_mm': [850,600,900,500,780,650,920,720,550,800,
                    880,620,760,940,580,820,700,960,540,790],
    'Temperature': [28,32,27,34,29,31,26,30,33,28,
                    27,32,29,26,33,28,30,25,34,29],
    'Fertilizer_kg': [60,45,65,40,55,48,70,52,42,58,
                      62,46,54,72,44,60,50,75,41,56],
    'Soil_Type': ['Loamy','Sandy','Loamy','Sandy','Clay','Sandy',
                  'Loamy','Clay','Sandy','Loamy','Loamy','Sandy',
                  'Clay','Loamy','Sandy','Clay','Clay','Loamy',
                  'Sandy','Clay'],
    'Crop_Yield_kg': [3200,2100,3500,1800,3000,2300,3700,2800,
                      1900,3100,3400,2200,2950,3800,2000,3150,
                      2700,3900,1850,3050]
}
df = pd.DataFrame(data)
print("Dataset Shape:", df.shape)
print("\nData Types:\n", df.dtypes)
print("\nMissing Values:\n", df.isnull().sum())
print("\nDuplicate Rows:", df.duplicated().sum())
print("\nStatistical Summary:\n", df.describe())
print("\nSoil Type Count:\n", df['Soil_Type'].value_counts())
sns.barplot(data=df, x='Soil_Type', y='Crop_Yield_kg')
plt.title("Average Crop Yield by Soil Type")
plt.show()
df['Crop_Yield_kg'].hist()
plt.title("Crop Yield Distribution")
plt.xlabel("Crop Yield (kg)")
plt.ylabel("Number of Farms")
plt.show()
sns.scatterplot(data=df, x='Rainfall_mm',
                y='Crop_Yield_kg', hue='Soil_Type')
plt.title("Rainfall vs Crop Yield")
plt.show()
sns.boxplot(data=df, x='Soil_Type', y='Crop_Yield_kg')
plt.title("Crop Yield by Soil Type")
plt.show()
```

```
Dataset Shape: (20, 5)

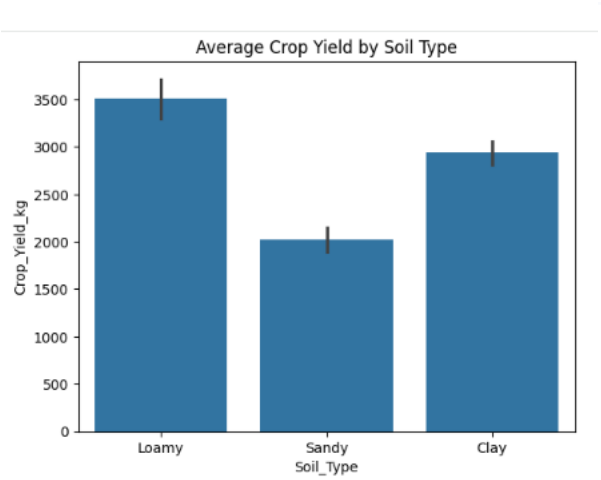
Data Types:
 Rainfall_mm       int64
Temperature        int64
Fertilizer_kg      int64
Soil_Type         object
Crop_Yield_kg      int64
dtype: object

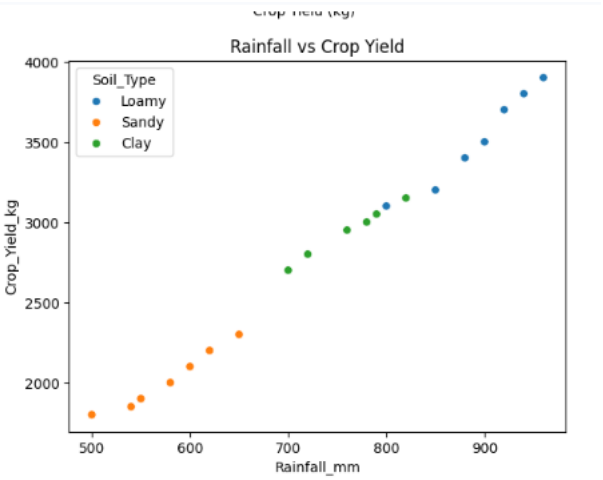
Missing Values:
 Rainfall_mm      0
Temperature       0
Fertilizer_kg     0
Soil_Type         0
Crop_Yield_kg     0
dtype: int64

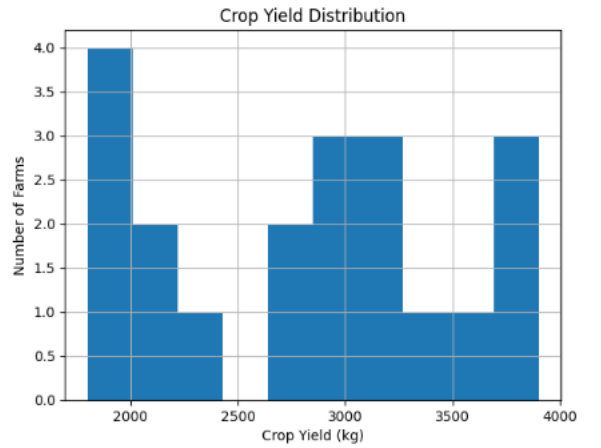
Duplicate Rows: 0

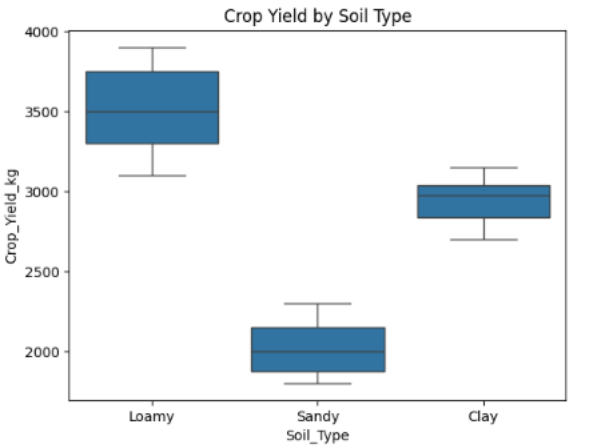
Statistical Summary:
        Rainfall_mm  Temperature  Fertilizer_kg  Crop_Yield_kg
count    20.000000    20.000000      20.000000      20.000000
mean    743.000000    29.550000      54.750000    2820.000000
std     144.335136     2.762055      10.507516     681.407213
min     500.000000    25.000000      40.000000    1800.000000
25%     615.000000    27.750000      45.750000    2175.000000
50%     770.000000    29.000000      54.500000    2975.000000
75%     857.500000    32.000000      60.500000    3250.000000
max     960.000000    34.000000      75.000000    3900.000000

Soil Type Count:
 Soil_Type
Loamy    7
Sandy    7
Clay     6
Name: count, dtype: int64
```









Common EDA Tasks:

1. Identify the number of rows and columns.
2. Check data types of each column.
3. Find missing values and duplicate values.
4. Calculate mean, median, minimum, maximum, and standard deviation.
5. Draw suitable charts such as bar chart, histogram, scatter plot, and box plot.
6. Identify important patterns from the dataset.
7. Write 4 to 5 observations based on EDA.
8. Give a short conclusion from the analysis.

RUBRICS FOR CALCULATION

Exploratory Data Analysis-Based Practical Assessment					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Dataset Understanding	20	Clearly explains dataset source, features, target variables, data types, and problem context with strong clarity	Explains dataset, features, and variables with adequate clarity	Basic explanation of dataset with limited understanding of variables	Poor understanding of dataset; variables and problem context are unclear
Data Cleaning	25	Effectively identifies and handles missing values, duplicates, incorrect entries, and outliers using suitable methods	Handles most missing values, duplicates, and errors correctly	Performs basic cleaning but misses some important data quality issues	Minimal or incorrect data cleaning; major errors remain in the dataset
Data Analysis and Statistical Summary	20	Provides detailed descriptive statistics, comparisons, patterns, and meaningful interpretations	Provides relevant summary statistics and basic interpretation	Provides limited statistics with weak explanation	Lacks statistical analysis or gives incorrect interpretation
Data Visualization	20	Uses highly appropriate charts with proper labels, titles, and clear visual interpretation	Uses suitable charts with mostly clear labels and explanation	Uses limited or basic charts with unclear interpretation	Poor or incorrect visualizations; charts do not support analysis
Insight Generation and Reporting	15	Presents strong insights, conclusions, and recommendations based on data evidence	Presents clear findings and conclusions with reasonable support	Provides basic findings but lacks depth or proper justification	Findings are unclear, unsupported, or not related to the data